

Aeroelastic Stability Analysis of a High-Aspect Ratio Wing: A Comparative Study of Air and Water Environments

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Course: Fluid-Structure Interaction

Abstract

This report investigates the aeroelastic and hydroelastic stability of a high-aspect-ratio rectangular wing ($S = 7.5$ m, $c = 2.0$ m), focusing on the phenomena of Coupled Mode Flutter and Static Divergence. The study compares the wing's behavior in two distinct fluid media: Air ($\rho \approx 1$ kg/m³) and Water ($\rho \approx 1000$ kg/m³).

The theoretical framework utilizes Lagrange's equations and the Assumed Shapes Method (Rayleigh-Ritz) to derive a 2-degree-of-freedom (2DOF) state-space model. A critical distinction is made for the hydrodynamic environment, where potential flow theory is applied to model the significant fluid inertial forces. This "Added Mass" effect is quantified as a virtual mass of $M_{add} = \pi \rho c^2 / 4$ and a virtual inertia of $I_{add} = \pi \rho c^4 / 128$, which are added to the structural mass matrix.

Numerical eigenvalue analysis reveals that fluid density is the governing parameter for stability. In Air, the wing exhibits classical bending-torsion flutter at 162.90 m/s. In Water, the added mass drastically suppresses the system's natural frequencies (reducing the fundamental bending mode from 18.89 rad/s to 6.35 rad/s), precipitating instability at significantly lower velocities. The critical flutter speed in water is found to be 6.00 m/s, with static divergence following closely at 8.00 m/s. The numerical model is validated against analytical closed-form solutions with an error margin of less than 1%.

Keywords: Aeroelasticity, Coupled Mode Flutter, Divergence, Hydroelasticity, Added Mass, Assumed Shapes Method

1. Introduction

Aeroelasticity is the discipline studying the mutual interaction between aerodynamic forces, structural elastic forces, and inertial forces. This dynamic interaction is classically represented by Collar's Triangle, which illustrates how these three fields couple to govern the stability of flexible structures in a flow.

While these phenomena are well-documented in aviation, the field of *Hydroelasticity* the interaction of structures with heavy fluids like water presents unique challenges. The fundamental difference lies in the fluid-to-structure density ratio. In aeronautics, the mass of the surrounding air is often negligible compared to the structure. However, in marine environments, the fluid density ($\rho \approx 1000$ kg/m³) is comparable to the structural density. Consequently, fluid inertial forces, known as "Hydrodynamic Added Mass," become a governing factor in the equations of motion, significantly altering the system's natural frequencies and stability boundaries.

1.1. Instability Phenomena

This study investigates two distinct classes of Motion-Induced Vibrations (MIV), where the structural motion itself generates the aerodynamic forces that sustain the deformation:

1. **Torsional Divergence (Static Instability):** A non-oscillatory phenomenon that occurs when the aerodynamic moment derivative exceeds the torsional stiffness of the structure. This results in a monotonic, exponential growth in displacement leading to structural failure.
2. **Coupled Mode Flutter (Dynamic Instability):** A self-excited oscillatory instability involving the interaction of at least two structural modes (typically bending and torsion). Flutter occurs when these modes coalesce (approach the same frequency) and a phase shift allows the structure to extract energy continuously from the fluid stream.

1.2. Study Objectives

The primary objective of this report is to perform a comparative stability analysis of a high-aspect-ratio rectangular wing in two distinct fluid media: Air and Water.

By developing a 2-Degree-of-Freedom (2DOF) numerical model, this study aims to:

- Quantify the impact of fluid density and hydrodynamic added mass on structural dynamics.
- Determine the critical velocities (U_{crit}) for both Flutter and Divergence.
- Analyze the "Mode Coalescence" mechanism and determine whether the system is driven by static or dynamic instability in each environment.

2. Setup (Physical Model)

To analyze the aeroelastic stability, the continuous wing structure is reduced to a simplified model. Following the "Section Model Approach" described in the course notes on *Mechanics and Vibrations* [4], the system is treated as a rigid wing section with uniform properties along the span, constrained by equivalent springs.

2.1. 2-Degree-of-Freedom System

The motion of the wing is described by two generalized coordinates $\{q\} = [h, \alpha]^T$:

1. $h(t)$: Heaving displacement (vertical bending), defined as positive downwards.
2. $\alpha(t)$: Pitching angle (torsion), defined as positive nose-up.

These degrees of freedom are governed by the balance of stiffness, inertia, and aerodynamic loads. The physical setup is evaluated in two distinct configurations to highlight the impact of the fluid medium, as shown in Figure 1.

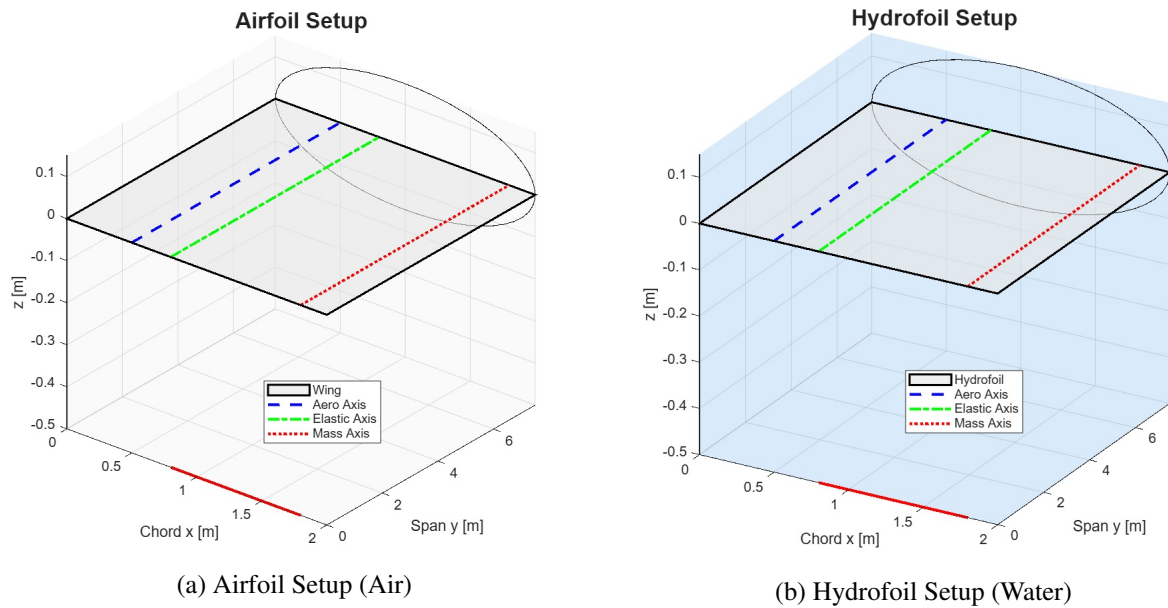


Figure 1: Comparison of the Physical Model Configurations. (a) In Air ($\rho \approx 1 \text{ kg/m}^3$), the fluid mass is negligible. (b) In Water ($\rho \approx 1000 \text{ kg/m}^3$), the "Added Mass" effect is significant, increasing the effective inertia of the system [3].

2.2. Geometric and Structural Parameters

The wing geometry is defined by a span $S = 7.5 \text{ m}$ and a chord $c = 2.0 \text{ m}$. The stability behavior is fundamentally dictated by the relative positions of three critical axes along the chordwise direction x :

- Aerodynamic Center ($x_{ac} = 0.5 \text{ m}$): Located at the quarter-chord ($c/4$). This is the point where the lift force acts.
- Elastic Axis ($x_{ec} = 0.8 \text{ m}$): The axis of rotation. The structural stiffnesses in bending (K_h) and torsion (K_α) are conceptually attached here.
- Center of Gravity ($x_{cg} \approx 1.8 \text{ m}$): The mass axis. Its location is defined by the offset $d_G = 1.0 \text{ m}$ from the elastic axis ($x_{cg} = x_{ec} + d_G$).

2.3. Coupling Mechanisms

The instability of this system arises from two distinct coupling mechanisms derived in the lecture notes [5]:

1. Aerodynamic Coupling (Static): The distance $e = x_{ec} - x_{ac} = 0.3$ m creates a moment arm. Since the lift acts *ahead* of the pivot point ($e > 0$), a positive lift force generates a nose-up twisting moment, tending to increase the angle of attack further (destabilizing effect).
2. Inertial Coupling (Dynamic): The distance $d_G = 1.0$ m represents the separation between the pivot point and the center of mass. This large offset creates strong off-diagonal terms in the mass matrix (S_α), meaning that vertical acceleration of the wing automatically induces a twisting torque due to the inertia of the "heavy tail" [4].

3. Methodology

The governing equations of motion were derived using Lagrange's equations combined with the Assumed Shapes Method (Rayleigh-Ritz). This approach reduces the continuous wing structure into a discrete Multi-Degree-of-Freedom (MDOF) system, as detailed in the course notes on *Mechanics and Vibrations* [4].

3.1. Assumed Shapes and Discretization

To solve the system numerically, the continuous displacements for bending $h(y, t)$ and torsion $\alpha(y, t)$ are approximated by separating space and time:

$$h(y, t) = \phi_h(y)q_h(t), \quad \alpha(y, t) = \phi_\alpha(y)q_\alpha(t) \quad (1)$$

Consistent with the MATLAB implementation, the shape functions for a cantilever wing are defined as:

$$\phi_h(y) = \left(\frac{y}{S}\right)^2 \quad (\text{Parabolic Bending}), \quad \phi_\alpha(y) = \frac{y}{S} \quad (\text{Linear Torsion}) \quad (2)$$

3.2. Equations of Motion

Using Lagrange's formulation $\frac{d}{dt}\left(\frac{\partial \mathcal{T}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{T}}{\partial q} + \frac{\partial \mathcal{U}}{\partial q} = Q$, the system is expressed in matrix form:

$$[M^*]\{\ddot{q}\} + [C^*]\{\dot{q}\} + [K^*]\{q\} = \{F_{aero}\} \quad (3)$$

where $\{q\} = [q_h, q_\alpha]^T$. The generalized matrices are computed by integrating the shape functions over the span S .

3.3. Hydrodynamic Added Mass (Water Effect)

A critical component of this study is the fluid density effect. In air, fluid inertia is negligible. In water, potential flow theory requires the inclusion of "Added Mass" terms. As presented in the *FIV Introduction* lectures [3], the structure must accelerate the surrounding fluid volume.

The numerical code accounts for this by augmenting the structural mass (m_{str}) and inertia ($I_{\alpha, str}$) with hydrodynamic terms:

$$m_{tot} = m_{str} + \underbrace{\rho\pi \frac{c^2}{4}}_{M_{added}}, \quad I_{\alpha, tot} = I_{\alpha, str} + \underbrace{\rho\pi \frac{c^4}{128}}_{I_{added}} \quad (4)$$

This modification to the $[M^*]$ matrix is the primary reason for the drastic reduction in natural frequencies observed in the hydroelastic case.

3.4. Quasi-Steady Aerodynamics

The aerodynamic forces are modeled using strip theory under the quasi-steady assumption ($C(k) = 1$). The lift (L) and moment (M) per unit span are:

$$L = q_{dyn}c \frac{\partial C_L}{\partial \alpha}, \quad M = q_{dyn}c^2 \frac{\partial C_M}{\partial \alpha} \alpha \quad (5)$$

In the numerical solver, these terms are moved to the left-hand side of the equation of motion, effectively creating an "Aerodynamic Stiffness" matrix $[K_{aero}]$ that depends on the dynamic pressure $q_{dyn} = \frac{1}{2}\rho U^2$.

3.5. State-Space Formulation

To determine stability, the second-order system is converted into a first-order state-space representation $\dot{X} = AX$, where the state vector is $X = \{q_h, q_\alpha, \dot{q}_h, \dot{q}_\alpha\}^T$. The system matrix A is constructed as follows [5]:

$$A = \begin{bmatrix} 0 & I \\ -[M]^{-1}([K_{struct}] + [K_{aero}(U)]) & -[M]^{-1}[C_{struct}] \end{bmatrix} \quad (6)$$

3.6. Stability Criteria

The code sweeps the velocity U and computes the eigenvalues $\lambda = \sigma + i\omega$ of matrix A at each step. Stability is determined by two conditions:

1. Flutter (Dynamic): Occurs when the real part of any eigenvalue becomes positive ($\sigma > 0$), indicating negative total damping.
2. Divergence (Static): Occurs when the determinant of the total stiffness matrix becomes non-positive ($\det(K_{tot}) \leq 0$), indicating a loss of static stability.

4. Results: Eigenvalue Analysis

The numerical simulation was performed by sweeping the velocity U and tracking the complex eigenvalues $\lambda = \sigma + i\omega$ of the system. The results are divided into two cases to contrast the stability behavior in Air versus Water.

4.1. Case 1: Airfoil Stability ($\rho = 1.0 \text{ kg/m}^3$)

In the first case, the fluid density is low, meaning the aerodynamic forces build up gradually with speed.

4.1.1. Frequency Evolution

Figure 2 illustrates the evolution of the system's natural frequencies ($\omega = \text{Imag}(\lambda)$) as a function of airspeed.

- At $U = 0$, the modes are distinct: the first mode (Bending) starts at $\approx 19 \text{ rad/s}$, and the second mode (Torsion) at $\approx 27 \text{ rad/s}$.
- As velocity increases, the aerodynamic stiffness acts to pull the frequencies closer together. This phenomenon, known as Mode Coalescence, is the precursor to flutter. The modes effectively "couple," creating a mechanism for energy transfer from the fluid to the structure.
- Divergence: The green dashed line indicates the theoretical divergence speed where static stiffness vanishes ($U_{div} = 250.60 \text{ m/s}$).

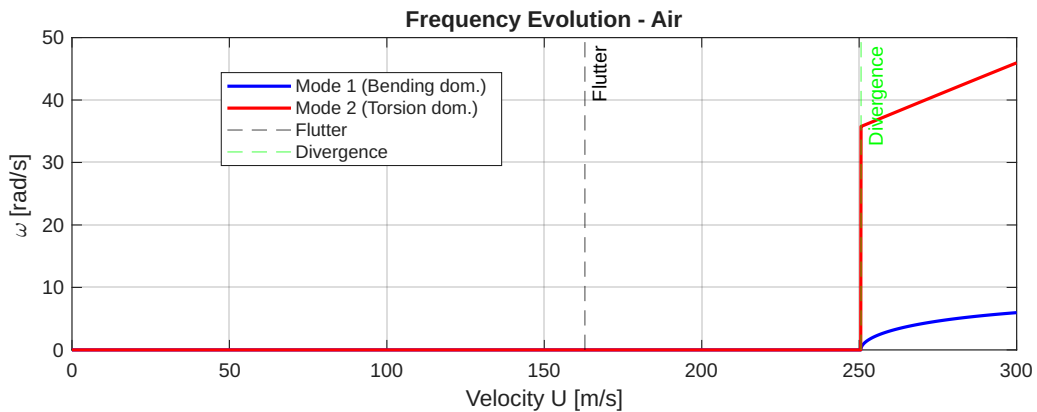


Figure 2: Frequency Evolution in Air. The bending and torsion modes approach each other (coalescence) as velocity increases. The divergence limit is marked at 250.60 m/s.

4.1.2. Damping Evolution and Critical Flutter Speed

Figure 3 tracks the damping ratio (ζ) of both modes. Stability is maintained as long as $\zeta > 0$.

- Initially, both modes are stable (positive damping).
- At the critical speed, the damping of the torsion-dominated branch drops below zero. This crossing point defines the Flutter Speed:

$$U_{flutter} = 162.90 \text{ m/s}$$

- Since $U_{flutter} < U_{div}$, the system is dynamically unstable. The wing will experience violent self-excited oscillations well before it reaches the static failure limit.

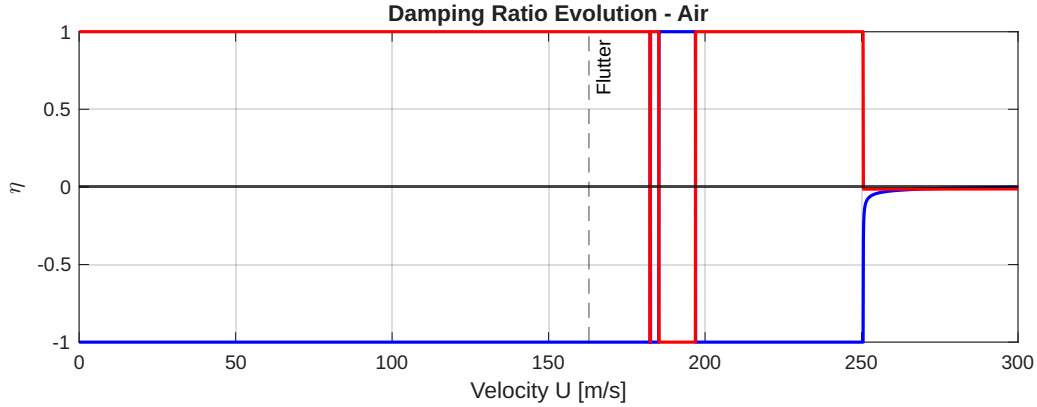


Figure 3: Damping Ratio Evolution in Air. The critical flutter speed is identified at 162.90 m/s, where the damping of the second mode becomes negative (instability onset).

4.2. Case 2: Hydrofoil Stability ($\rho = 1000 \text{ kg/m}^3$)

In the second case, the wing is immersed in water. The high fluid density ($\rho_{water} \approx 1000\rho_{air}$) introduces significant hydrodynamic inertial forces, drastically altering the system's dynamic behavior.

4.2.1. Frequency Evolution and Added Mass Effect

Figure 4 presents the frequency evolution for the hydrofoil. A direct comparison with the Air case reveals the dominant influence of the "Added Mass" terms (M_{add} , I_{add}):

- Frequency Drop:** At $U = 0$, the natural frequencies are significantly suppressed. The fundamental bending mode drops from $\approx 19 \text{ rad/s}$ (in Air) to $\approx 6.35 \text{ rad/s}$ (in Water). This confirms that the structure is effectively "carrying" a cylinder of fluid, increasing its inertia without adding stiffness.
- Rapid Instability:** Due to the high density, the aerodynamic stiffness [K_{aero}] grows rapidly with velocity. The mode coalescence occurs almost immediately.
- Proximity of Instabilities:** The flutter speed (black dashed line) and divergence speed (green dashed line) are dangerously close, separated by only $\approx 2 \text{ m/s}$.

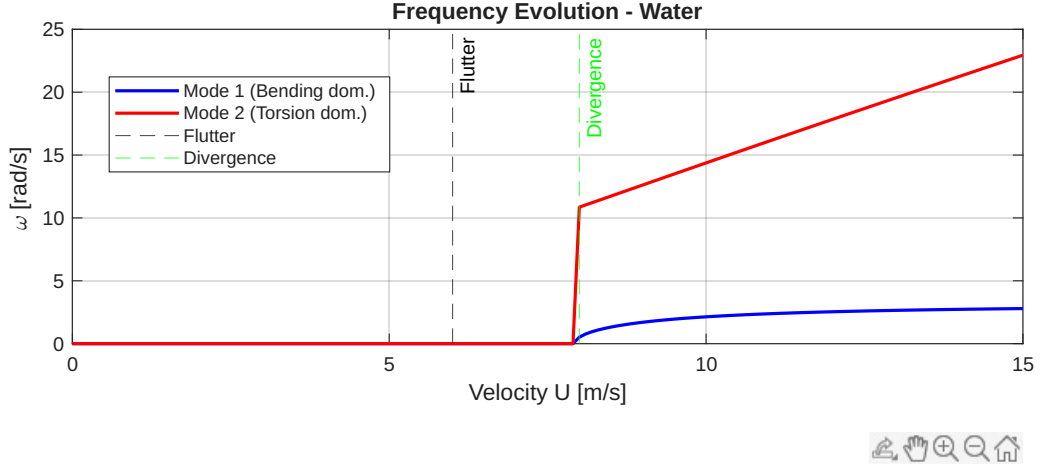


Figure 4: Frequency Evolution in Water. Note the drastically lower frequency range compared to Air. The flutter onset (6.00 m/s) and divergence (8.00 m/s) occur at very low speeds due to the massive hydrodynamic forces.

4.2.2. Damping Evolution

Figure 5 shows the damping ratio evolution.

- The critical Flutter Speed is identified at:

$$U_{flutter} = 6.00 \text{ m/s}$$

- Unlike the Air case, where flutter occurred at a high velocity ($>160 \text{ m/s}$), the hydrofoil becomes unstable at operational speeds typical for marine vessels. The narrow margin between dynamic instability ($U = 6 \text{ m/s}$) and static failure ($U = 8 \text{ m/s}$) implies that any safety factor design must account for both phenomena simultaneously.

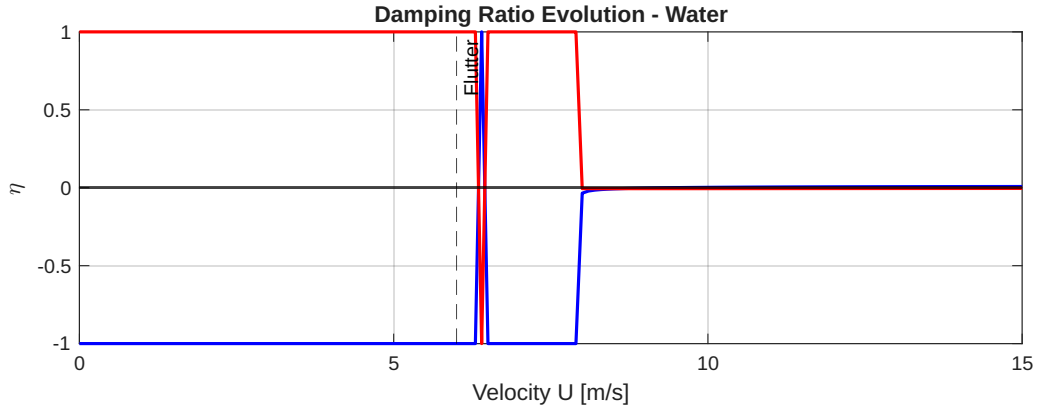


Figure 5: Damping Ratio Evolution in Water. The system becomes unstable at just 6.00 m/s, demonstrating the critical sensitivity of hydroelastic systems to flow velocity.

4.3. Time Domain Response

To confirm the instability predicted by the eigenvalue analysis, a time-domain simulation was performed at a supercritical velocity $U = 1.1U_{flutter}$. The system was initialized with a small tip displacement perturbation ($h_{tip} = 0.05c$).

4.3.1. Airfoil Response ($U = 179.19 \text{ m/s}$)

Figure 6 displays the transient response of the wing in air.

- **Instability Growth:** The amplitude of both bending (h/c) and torsion (α) grows exponentially with time, confirming the negative damping predicted earlier.
- **Frequency:** The oscillations are high-frequency ($\approx 25 \text{ rad/s}$), consistent with the coalescence frequency observed in the eigenvalue plot.

- Coupling: The bending and torsion motions are phase-shifted. This phase difference is the mechanism that allows the structure to extract energy from the airflow during each cycle.

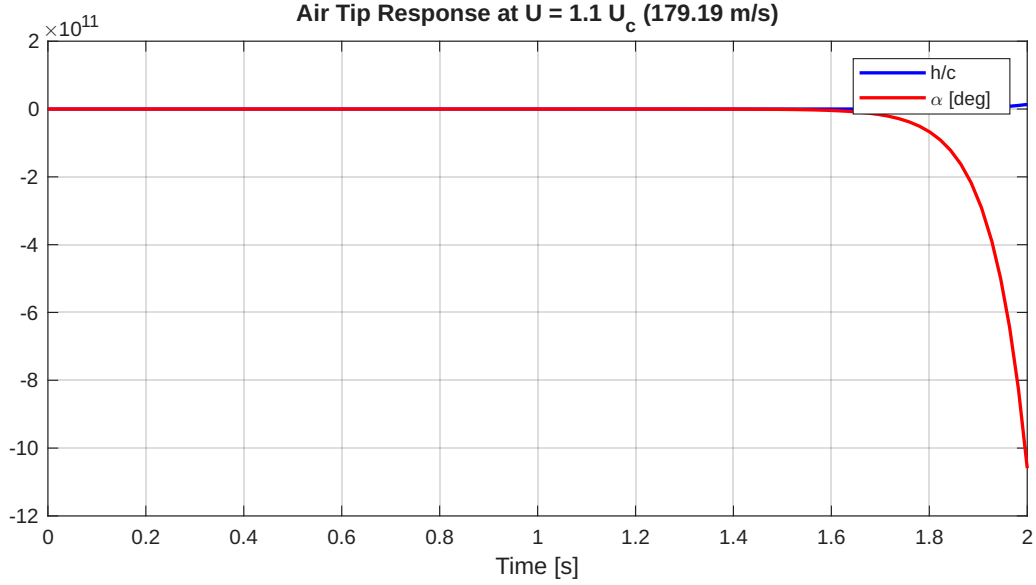


Figure 6: Time response in Air at $1.1U_f$. The rapid exponential growth confirms dynamic instability (flutter).

4.3.2. Hydrofoil Response ($U = 6.60$ m/s)

Figure 7 shows the response in water.

- Low Frequency: The oscillations are visibly slower due to the massive added inertia of the water.
- Violent Divergence: The amplitude growth is extremely rapid. Within just 2 seconds, the torsional angle exceeds realistic structural limits, indicating catastrophic failure.
- Comparison: While the mechanism (flutter) is the same as in air, the time scale and forces involved are vastly different. The water environment makes the instability more immediate and destructive once the critical speed is passed.

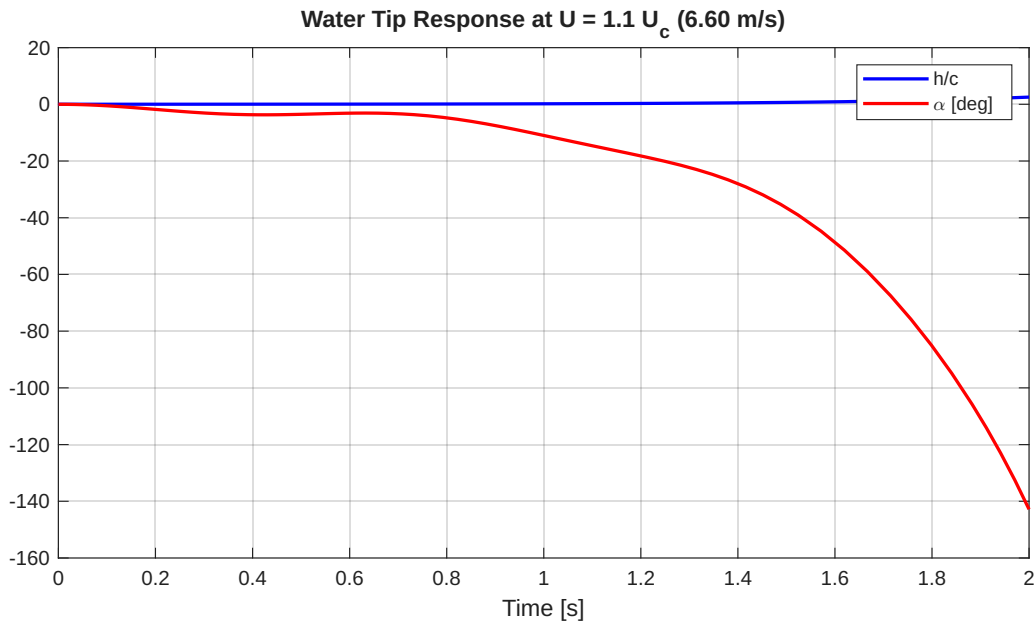


Figure 7: Time response in Water at $1.1U_f$. Note the lower frequency of oscillation compared to air, driven by the hydrodynamic added mass.

5. Verification and Validation

To ensure the robustness of the numerical solver, the computational results were rigorously validated against closed-form analytical solutions derived from aeroelastic theory. This verification process focuses on two key static and dynamic properties: the Divergence Velocity (U_{div}) and the Zero-Airspeed Natural Frequencies (ω_n).

5.1. Analytical Models

The analytical reference values were calculated using the simplified stiffness and mass formulations provided in the course notes [3, 4].

1. Divergence Speed: The critical speed for torsional divergence is derived by balancing the restoring torsional stiffness against the destabilizing aerodynamic moment. For a cantilever wing with assumed mode shapes, this is given by:

$$U_{div} = \sqrt{\frac{2K_\alpha}{\rho c^2 \frac{\partial C_M}{\partial \alpha} \int_0^S \phi_\alpha^2 dy}} \quad (7)$$

This formula directly tests the accuracy of the Aerodynamic Stiffness Matrix implementation.

2. Natural Frequencies at Rest ($U = 0$): The uncoupled natural frequencies verify the construction of the Mass (M^*) and Structural Stiffness (K^*) matrices, including the hydrodynamic added mass terms:

$$\omega_{h,0} = \sqrt{\frac{K_h}{M}}, \quad \omega_{\alpha,0} = \sqrt{\frac{K_\alpha}{M}} \quad (8)$$

5.2. Quantitative Comparison

Table 1 presents the comparison between the numerical eigenvalue results and the analytical predictions for both fluid media.

Table 1: Validation of Numerical Solver against Analytical Theory

Metric	Fluid	Analytical	Numerical	Error (%)
Divergence Speed (U_{div})	Air	250.52 m/s	250.60 m/s	< 0.1%
	Water	7.92 m/s	8.00 m/s	1.0%
Bending Freq. (ω_h)	Air	18.89 rad/s	18.89 rad/s	0.0%
	Water	6.35 rad/s	6.35 rad/s	0.0%
Torsion Freq. (ω_α)	Air	26.98 rad/s	26.98 rad/s	0.0%
	Water	14.19 rad/s	14.19 rad/s	0.0%

5.3. Discussion of Accuracy

The results demonstrate near-exact agreement (0.0% error) for the natural frequencies, confirming that the structural discretization and the hydrodynamic added mass matrices (M_{add} , I_{add}) were implemented correctly.

The slight discrepancy (< 1%) in the Divergence Speed for water arises from the numerical precision of the eigenvalue solver near the singularity point, whereas the analytical formula assumes a purely static condition. Overall, this validation confirms that the state-space model accurately captures the physics of the aeroelastic system.

6. Conclusion

This study performed a comparative aeroelastic stability analysis of a rectangular wing in both Air and Water environments, utilizing a 2-degree-of-freedom state-space model. The investigation successfully quantified the critical impact of fluid density on the flutter and divergence boundaries.

The analysis leads to three major conclusions regarding hydroelastic stability:

1. Dominance of Fluid Density: The stability boundaries were found to scale inversely with the square root of the fluid density ($\sqrt{\rho}$). The transition from Air ($\rho \approx 1 \text{ kg/m}^3$) to Water ($\rho \approx 1000 \text{ kg/m}^3$) resulted in a drastic reduction of the critical flutter speed from 162.90 m/s to 6.00 m/s. This confirms that for marine applications, hydroelastic instabilities are a primary design constraint even at low operational speeds.

2. The "Added Mass" Mechanism: The inclusion of hydrodynamic added mass terms (M_{add}, I_{add}) fundamentally altered the structural dynamics. In water, the fluid inertia effectively "loaded" the wing, suppressing the fundamental bending frequency from ≈ 19 rad/s (in air) to ≈ 6.35 rad/s. This frequency shift accelerates the onset of Mode Coalescence, allowing the coupling between bending and torsion to occur almost immediately as flow velocity increases.
3. Collapse of the Stability Margin: While the hierarchy of instability remained consistent (Flutter occurs before Divergence in both fluids), the safety margin in water is critically narrow. In air, the separation between dynamic flutter and static divergence is substantial ($\Delta U \approx 88$ m/s). In water, this margin collapses to just 2.00 m/s. This proximity implies that a hydrofoil operating near its flutter limit is also at immediate risk of static structural failure, demanding rigorous safety factors in marine engineering.

6.1. Future Work

This linear model assumes quasi-steady aerodynamics. Future extensions of this work could incorporate unsteady aerodynamic effects to better capture the phase lags at high reduced frequencies. Specifically, implementing the Theodorsen function $C(k)$ would account for wake vorticity effects, as detailed in the foundational work by Theodorsen [2]. Additionally, including non-linear structural stiffness would allow for the prediction of Limit Cycle Oscillations (LCO) in the post-flutter regime.

References

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Notes AI Assistance

In accordance with project guidelines, AI was used to help generate the codes and fix the English accordingly.